GUMBEL MDA & TAIL BEHAVIOUR OF INFIMA OF GAUSSIAN PROCESSES

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ABSTRACT. We study exact high-level asymptotics for probabilities of the form

$$\mathbb{P}\left\{\inf_{t\in\mathcal{T}} X(t) > u\right\}, \qquad u\to\infty,$$

where X is a centred Gaussian process (GP) with continuous sample paths on a nonempty compact parameter space $\mathcal{T}$. As suggested by the findings of [1], the exponential rate is governed by an optimal-measure problem, while the exact prefactor depends on the local geometry of the contact set. The main results treat finite-contact regimes in which the optimal measure has finite support. We separate dominated contacts, where residual fluctuations are negligible, from smooth-critical contacts, where a residual derivative gives an explicit quadratic correction, and rough-critical contacts, where a tangent GP contributes the local correction. We also study full-contact regimes through a Laplace-transform reduction and obtain logarithmic refinements for rough fBM. Applications include integrated fractional Brownian motion, a two-interval fractional Brownian phase transition, a fractional Brownian sheet, passive contacts, and smooth and rough endpoint contacts.

1. INTRODUCTION

Let $X(t)$, $t\in\mathcal{T}$ be a centred GP with continuous sample paths on a nonempty compact metric space $\mathcal{T}$ and let $R(s,t), s,t\in\mathcal{T}$ be its covariance function. Motivated by [1], we study the extreme value properties of the infimum $\mathfrak{m}_X = \inf_{t\in\mathcal{T}} X(t)$. In order to avoid degenerate cases, we shall assume hereafter that there exists $u_0>0$ such that for all $u>u_0$

$$p_{\mathcal{T}}(u) := \mathbb{P}\{\mathfrak{m}_X > u\} > 0. \tag{1}$$

For every nonempty compact $A\subseteq\mathcal{T}$ write $p_A(u) := \mathbb{P}\{\inf_{t\in A} X(t) > u\}$.
We consider first the simplest case $\mathcal{T}=\{t_1,\dots,t_n\}, n\geq 1$ with $\Sigma = (R(t_i,t_j))_{1\leq i,j\leq n}$ nonsingular. It is well-known that the asymptotics of $p_{\mathcal{T}}(u)$ as $u\to\infty$ are governed by the quadratic programming problem

$$\min_{\boldsymbol{p}\in\Delta_n} \boldsymbol{p}^\top\Sigma\boldsymbol{p}, \qquad \Delta_n := \left\{\boldsymbol{p}\in[0,\infty)^n : \sum_{i=1}^n p_i = 1\right\}$$

see [2]. Let $\boldsymbol{p}_*$ be the minimiser and put

$$\sigma_*^2 := (\boldsymbol{p}_*)^\top\Sigma\boldsymbol{p}_* > 0, \qquad m(t_i) := \frac{(\Sigma\boldsymbol{p}_*)_i}{\sigma_*^2}, \quad i=1,\dots,n.$$

The optimality conditions imply that

$$m(t_i)\geq 1, \qquad i=1,\dots,n$$

with equality whenever the ith component of $\boldsymbol{p}_*$ satisfies $p_{*,i}>0$. Thus if

$$S := \{t_i\in\mathcal{T} : p_{*,i}>0\}$$

is the active set, $E=\{t_i\in\mathcal{T} : m(t_i)=1\}$ is the contact set, and $P=E\setminus S$ is the passive set with ℓ elements, then

$$p_{\mathcal{T}}(u)\sim p_E(u)\sim C\mathbb{P}\{\min_{t\in S} X(t) > u\}, \qquad u\to\infty, \tag{2}$$

where $C=1$ if $\ell=0$, while for $\ell>0$, C is the orthant probability of the centred Gaussian regression residual at the passive points $t_i\in P$; see Lemma 4.2 and [2, Corollary 4.1] and [3, Example 6.1 and Theorem 5.1]. Moreover, [4, Theorem 2.1] implies that the distribution function F of $\mathfrak{m}_X$ belongs to the Gumbel max-domain of attraction (GMDA) with scaling function a satisfying $a(u)\sim\sigma_*^2/u$ as $u\to\infty$, i.e.,

$$\lim_{u\to\infty}\frac{1-F(u+xa(u))}{1-F(u)} = e^{-x}, \quad \forall x\in\mathbb{R}. \tag{3}$$

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For $\mathcal{T}$ a compact interval of $\mathbb{R}$, the GMDA of F was established under smoothness assumptions in [1]. For continuous GPs on general compact metric spaces, the GMDA property under the standing non-degeneracy assumption (1) follows from [4, Theorem 2.1].
When F has an eventually positive differentiable density F', define its reciprocal hazard by $a(u) := (1 - F(u))/F'(u)$ for all sufficiently large u. The additional condition

$$\lim_{u\to\infty} a'(u) = 0 \tag{4}$$

i.e., the von Mises condition, does not follow from GMDA and absolute continuity alone.
For a general compact parameter space $\mathcal{T}$ our first question concerns the stronger von Mises condition:

Q1. *Under what additional conditions does the reciprocal hazard of F satisfy the von Mises condition (4)?*

Next, let $\mathcal{M}_1(\mathcal{T})$ denote the set of Borel probability measures on $\mathcal{T}$. For every $\mu \in \mathcal{M}_1(\mathcal{T})$ averaging implies the following upper bound

$$p_{\mathcal{T}}(u) \le \mathbb{P}\{X_\mu > u\}, \qquad X_\mu := \int_{\mathcal{T}} X(t)\,\mu(dt). \tag{5}$$

By weak compactness of $\mathcal{M}_1(\mathcal{T})$ and continuity of the covariance function of X, the energy functional $\mu \mapsto \mathrm{Var}(X_\mu)$ is continuous under weak convergence of probability measures and the minimum is attained. Any minimiser $\mu_* \in \mathcal{M}_1(\mathcal{T})$ is called an *optimal measure* and we set

$$\sigma_*^2 := \min_{\mu\in\mathcal{M}_1(\mathcal{T})} \mathrm{Var}(X_\mu).$$

The non-degeneracy assumption (1) implies

$$\sigma_*^2 = \mathrm{Var}(X_{\mu_*}) > 0. \tag{6}$$

Indeed, if $\sigma_*^2 = 0$, then $X_{\mu_*} = 0$ almost surely, which is incompatible with $p_{\mathcal{T}}(u) > 0$ for $u > u_0$.
Next, let μ_* be an optimal measure. In view of the optimality criterion in Lemma A.1, its normalised potential

$$m(t) := \frac{\mathrm{Cov}(X(t), X_{\mu_*})}{\mathrm{Var}(X_{\mu_*})}, \qquad t \in \mathcal{T},$$

satisfies

$$m(t) \ge 1, \qquad t \in \mathcal{T}, \qquad \text{and} \qquad m(t) = 1 \qquad \text{on } \mathrm{supp}(\mu_*) =: S$$

and hence the support S, which may depend on the choice of optimal measure is contained in the contact set i.e.,

$$S \subset E := \{t \in \mathcal{T} : m(t) = 1\}.$$

The normalised potential m and hence E do not depend on the choice of optimal measure. Indeed, if μ and ν are optimal, then

$$\mathrm{Var}\left(\frac{X_\mu + X_\nu}{2}\right) = \sigma_*^2 - \frac{1}{4}\mathrm{Var}(X_\mu - X_\nu).$$

Since $(\mu + \nu)/2$ is a probability measure, optimality forces $X_\mu = X_\nu$ almost surely, so their potentials coincide. Moreover, both E and S are compact and further

$$X(t) = m(t)X_{\mu_*} + Z_*(t), \qquad t \in \mathcal{T}, \tag{7}$$

where the residual process Z_* is a centred GP independent of $X_{\mu_*} \sim N(0, \sigma_*^2)$, see e.g., [1, 4].
Thus S, E, and μ_* are the continuous counterparts of the active set, contact set, and optimal weights in the finite-dimensional problem introduced above.
The logarithmic asymptotic follows directly from (7). Put $Y := X_{\mu_*}$. Since $m \ge 1$, for every fixed $c > 0$ and $u > 0$, independence gives

$$\mathbb{P}\{Y > u + c\}\,\mathbb{P}\{\|Z_*\|_\infty < c\} \le p_{\mathcal{T}}(u) \le p_S(u) \le \mathbb{P}\{Y > u\}.$$

The support of a centred Gaussian measure on the separable Banach space $C(\mathcal{T})$ is a closed linear subspace and hence the small-ball factor is positive. Thus $\sigma_*^2 > 0$ also implies (1) and Gaussian tail asymptotics yield

$$\lim_{u\to\infty} \frac{1}{u^2}\log p_{\mathcal{T}}(u) = \lim_{u\to\infty} \frac{1}{u^2}\log p_S(u) = \lim_{u\to\infty} \frac{1}{u^2}\log \mathbb{P}\{X_{\mu_*} > u\} = -\frac{1}{2\sigma_*^2} < 0. \tag{8}$$

In this Gaussian setting log-concavity of $p_{\mathcal{T}}$ together with (8) yields the GMDA property, see [4, Section 3.2]. In the case $E = \mathcal{T}$, hereafter referred to as the *full-contact regime* even without knowing the support S explicitly (7) reduces to

$$X(t) = X_{\mu_*} + Z_*(t), \qquad t \in \mathcal{T}, \tag{9}$$

with Z_* a centred GP independent of $X_{\mu_*} \sim N(0, \sigma_*^2)$.
This class of GPs with full contact is broad. Indeed, if Z_* is a centred GP with continuous sample paths and if $\mu_* \in \mathcal{M}_1(\mathcal{T})$ satisfies

$$\int_{\mathcal{T}} Z_*(t)\, \mu_*(dt) = 0$$

almost surely, then for an independent Gaussian random variable (rv) $Y \sim N(0, \sigma_*^2)$ the process

$$X(t) := Y + Z_*(t), \qquad t \in \mathcal{T}$$

has μ_* as an optimal measure and has full contact set $E = \mathcal{T}$.
In the full-contact regime, integrating (9) against μ_* yields almost surely

$$\int_{\mathcal{T}} Z_*(t)\, \mu_*(dt) = 0 \tag{10}$$

and hence

$$\mathfrak{m}_{Z_*} := \inf_{t \in \mathcal{T}} Z_*(t) \le 0.$$

Moreover with $X_{\mu_*} \sim N(0, \sigma_*^2)$ independent of $\mathfrak{m}_{Z_*}$

$$\mathfrak{m}_X = X_{\mu_*} + \mathfrak{m}_{Z_*}$$

and consequently as shown already in [5] for every $u \in \mathbb{R}$

$$p_{\mathcal{T}}(u) = \frac{\exp\left(-u^2/(2\sigma_*^2)\right)}{\sqrt{2\pi}\sigma_*} \int_0^\infty \exp\left(-\frac{u\varepsilon}{\sigma_*^2} - \frac{\varepsilon^2}{2\sigma_*^2}\right) \mathbb{P}\{\mathfrak{m}_{Z_*} > -\varepsilon\}\, d\varepsilon. \tag{11}$$

An immediate Abelian consequence of the above identity is

$$\lim_{u \to \infty} \frac{p_{\mathcal{T}}(u)}{\mathbb{P}\{X_{\mu_*} > u\}} = \mathbb{P}\left\{\mathfrak{m}_{Z_*} = 0\right\}.$$

If $\mathbb{P}\left\{\mathfrak{m}_{Z_*} = 0\right\}$ is positive, then both the exact asymptotics of $p_{\mathcal{T}}(u)$ and the von Mises condition (4) hold.
We shall show that (4) is still valid even when $\mathbb{P}\left\{\mathfrak{m}_{Z_*} = 0\right\} = 0$. For the latter case, the exact formula (11) can be used to transfer the problem to the investigation of $r(\varepsilon) := \mathbb{P}\{\mathfrak{m}_{Z_*} > -\varepsilon\}$ as $\varepsilon \downarrow 0$. Indeed, if for some ultimately positive function $\mathcal{L}$, slowly varying at infinity

$$r(\varepsilon) \sim K\varepsilon^\alpha \mathcal{L}(1/\varepsilon), \qquad K > 0, \quad \alpha \ge 0 \tag{12}$$

holds as $\varepsilon \downarrow 0$, then by Karamata's Abelian theorem for Laplace transforms and (11) we obtain

$$p_{\mathcal{T}}(u) \sim \frac{K\Gamma(\alpha+1)\sigma_*^{2\alpha+1}}{\sqrt{2\pi}} u^{-(\alpha+1)} \mathcal{L}(u) \exp\left\{-\frac{u^2}{2\sigma_*^2}\right\}, \qquad u \to \infty. \tag{13}$$

Explicit calculations of the asymptotics of $r(\varepsilon)$, as $\varepsilon \downarrow 0$ are possible in a few particular cases, see Example 2.1 below.
Exact asymptotics of $p_{\mathcal{T}}(u)$ are available in particular in the smooth settings of [1] and in special Brownian cases; see [6, 7]. This leads to our second question:
Q2. *Under what conditions can we derive explicit exact leading asymptotics of $p_{\mathcal{T}}(u)$ as $u \to \infty$ in finite-contact regimes?*

We shall address the above question by restricting to the case where the contact set E is finite. This implies that every optimal measure μ_* has finite support and can be written as

$$\mu_* = \sum_{i=1}^k p_{*,i}\delta_{t_i}, \qquad p_{*,i} > 0, \qquad i = 1, \dots, k, \qquad \sum_{i=1}^k p_{*,i} = 1, \tag{14}$$

where δ_{t_i} denotes the Dirac probability measure at t_i.
The support is $S = \{t_1, \dots, t_k\}$ and the weight vector is $\boldsymbol{p}_* = (p_{*,1}, \dots, p_{*,k})^\top$. Writting $\Sigma_* = (R(t_i, t_j))_{1 \le i,j \le k}$ and $\boldsymbol{X}_S = (X(t_1), \dots, X(t_k))^\top$ Lemma A.1 below yields

$$\Sigma_* \boldsymbol{p}_* = \sigma_*^2 \mathbf{1}, \qquad \mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^k.$$

If Σ_* is nonsingular, Gaussian regression therefore implies

$$m(t)=\boldsymbol{a}(t)^\top\mathbf{1}\geq 1,\qquad X(t)=\boldsymbol{a}(t)^\top\boldsymbol{X}_S+Z(t),\qquad \boldsymbol{a}(t):=\Sigma_*^{-1}(R(t_1,t),\dots,R(t_k,t))^\top,\quad t\in\mathcal{T}, \tag{15}$$

where Z is a centred GP independent of $\boldsymbol{X}_S$. The support points of the optimal measure are special since

$$m(t_i)=1,\qquad Z(t_i)=0,\qquad 1\leq i\leq k.$$

The support points in S are called *active* contacts, while the points in $P=E\setminus S$ are called *passive* contacts. In general the two residual processes Z and Z_* differ since

$$Z_*(t)-Z(t)=\boldsymbol{a}(t)^\top\big(\boldsymbol{X}_S-\mathbf{1}X_{\mu_*}\big),\qquad t\in\mathcal{T},$$

however if S is a singleton, then

$$Z(t)=Z_*(t),\qquad t\in\mathcal{T}.$$

The reduction in (2) to a positive constant times $p_S(u)$ need not hold for an infinite parameter set, even when E is finite. To see this, let ξ,η be independent standard normal rvs and for $\alpha,\lambda>0$ set

$$X(t):=\xi\big(1+\lambda|t|^\alpha\big)+\eta t,\qquad t\in\mathcal{T}=[-1,1].$$

For this process, we have

$$\mu_*=\delta_0,\qquad S=E=\{0\},\qquad \sigma_*^2=1,\qquad m(t)-1=\lambda|t|^\alpha,\qquad Z(t)=\eta t,\qquad t\in[-1,1]$$

and the residual GP Z satisfies the mean-square Hölder condition

$$\mathrm{Var}(Z(t+h)-Z(t+h'))\leq c_H|h-h'|^{2\beta},\qquad t,t+h,t+h'\in\mathcal{T},$$

with $\beta=c_H=1$. Direct calculation gives

$$\lim_{u\to\infty}\frac{p_{\mathcal{T}}(u)}{\mathbb{P}\{\xi>u\}}=\begin{cases}1, & 0<\alpha<2,\\ \left(1+\dfrac{1}{2\lambda}\right)^{-1/2}, & \alpha=2,\\ 0, & \alpha>2.\end{cases}$$

The three regimes in the above limit reflect the competition between the rate at which $m(t)-1$ vanishes near 0, controlled by α, and the residual smoothness exponent $\beta=1$.
The cases $\alpha<2$ and $\alpha=2$ are covered by Theorem 3.1 and the interior version in Remark 3.7, respectively. The regime in which the ratio vanishes, illustrated by $\alpha>2$, is outside the scope of our general exact-asymptotic results.
Critical contacts balance potential separation of order $|h|^{2\beta}$ with residual fluctuations on the scale $|h|^\beta$. The *smooth-critical* case has quadratic separation and a linear residual tangent, as when $\alpha=2$ above. Critical contacts with $\beta\in(0,1)$ are called *rough-critical*. The Brownian endpoint model in Example 3.8 has $\beta=1/2$ and gives an explicit nontrivial correction factor.
Passive orthant factors and smooth quadratic corrections already appear under stronger smoothness assumptions in [1, Theorem 4.1 and equation (4.3)]. Our finite-contact results use local assumptions and also treat rough tangents.
A further motivation comes from [8], where the expected-infimum and persistence asymptotics of Brown–Resnick stationary log-normal processes exhibit endpoint domination for fractional Brownian motion with $H>1/2$, in contrast to the Brownian boundary case. Those results concern drifted processes on growing intervals. Here we study high minima on fixed compact sets and explain the corresponding distinction through the geometry of the contact set.
Our main contributions are as follows:

(1) We give an exact finite-contact reduction for dominated support contacts, including the Gaussian orthant factor generated by finitely many passive contacts. Our results cover also general random fields including the fractional Brownian sheet and several other models.
(2) For a one-point active endpoint, we identify the exact corrections in rough-critical and smooth-critical regimes through a tangent process or a residual derivative.
(3) In the full-contact regime, we establish the canonical von Mises condition and derive logarithmic refinements for rough fBM using the Laplace-transform representation and small-deviation estimates.

The von Mises condition is established in the full-contact regime; its validity under the general finite-contact assumptions is not addressed here.
Short organisation of the paper: Section 2 treats the full-contact regime. Section 3 contains the finite-contact results and examples. The proofs are collected in Section 4, and the Appendix records the auxiliary optimal-measure results used in the main text.

2. Full-Contact Regime

Let $X(t), t \in \mathcal{T}$, $\mu_*, \sigma_*^2, S, E, m(t), t \in \mathcal{T}$ be as in the Introduction and keep our standing assumption $\sigma_*^2 := \mathrm{Var}(X_{\mu_*}) > 0$.
Except for the comparison in the opening example, this section concerns the full-contact regime

$$E = \{t \in \mathcal{T} : m(t) = 1\} = \mathcal{T}.$$

The transfer principle in (9) allows us to reduce the problem of the tail asymptotics of $p_{\mathcal{T}}(u)$ to the study of the small deviation asymptotics of the residual process $Z_*(t), t \in \mathcal{T}$. In particular cases this is a powerful tool for Q1 and for exact full-contact asymptotics. The following fractional Brownian example compares this regime with dominated contact. Throughout the paper we denote by B_H, $H \in (0,1)$ a standard fractional Brownian motion (fBM).

Example 2.1. Let B_H have Hurst index $H \in [1/2, 1)$, fix $0 < a_1 < b_1 < a_2 < b_2 < \infty$, and set

$$\mathcal{K} := [a_1, b_1] \cup [a_2, b_2], \qquad p_H(u) := \mathbb{P}\left\{\inf_{t \in \mathcal{K}} B_H(t) > u\right\}.$$

Suppose first that $H > 1/2$, then the normalised potential of δ_{a_1} is

$$m_H(t) = \frac{t^{2H} + a_1^{2H} - (t - a_1)^{2H}}{2a_1^{2H}}, \qquad t \in \mathcal{K}.$$

Since

$$t^{2H} > a_1^{2H} + (t - a_1)^{2H}, \qquad t > a_1,$$

Corollary A.2 implies that the unique optimal measure is

$$\mu_* = \delta_{a_1}, \qquad S = E = \{a_1\}, \qquad P = \varnothing.$$

Moreover we have

$$m_H(t) - 1 \sim \frac{H}{a_1}(t - a_1), \qquad t \downarrow a_1.$$

For $Z(t) := B_H(t) - m_H(t) B_H(a_1)$, the regression-residual bound in Remark 3.2 gives

$$\mathbb{E}\{\big(Z(t) - Z(s)\big)^2\} \le |t - s|^{2H}, \qquad s, t \in \mathcal{K}.$$

Thus the dominated-contact assumptions hold with $q = 1$ and $\beta = H$. Theorem 3.1 therefore yields

$$p_H(u) \sim \mathbb{P}\{B_H(a_1) > u\} \sim \frac{a_1^H}{\sqrt{2\pi}\, u} \exp\left\{-\frac{u^2}{2a_1^{2H}}\right\}, \qquad u \to \infty.$$

Now let $H = 1/2$ and write $B := B_{1/2}$. In this case $m_{1/2} \equiv 1$ on $\mathcal{K}$, so $E = \mathcal{K}$. The measure δ_{a_1} is nevertheless the unique optimal measure, because for every $\mu \in \mathcal{M}_1(\mathcal{K})$

$$\mathrm{Var}\left(\int_{\mathcal{K}} B(t)\, \mu(dt)\right) = a_1 + \int_{a_1}^{b_2} \mu([x, b_2])^2\, dx \ge a_1,$$

with equality only for $\mu = \delta_{a_1}$ and hence

$$\sigma_*^2 = a_1, \qquad Z_*(t) = B(t) - B(a_1).$$

Set now

$$A := b_1 - a_1, \qquad G := a_2 - b_1, \qquad C := b_2 - a_2,$$

and let

$$W(t) := B(a_1 + t) - B(a_1), \qquad t \ge 0.$$

Then W is a standard Brownian motion independent of $B(a_1)$. Writing

$$\varphi_t(x) := \frac{1}{\sqrt{2\pi t}} \exp\left\{-\frac{x^2}{2t}\right\}, \qquad \Phi(x) := \int_{-\infty}^{x} \varphi_1(y)\, dy$$

the reflection principle on the first interval and the Markov property across the gap yield

$$\begin{aligned} r_{\mathcal{K}}(\varepsilon) &:= \mathbb{P}\left\{\inf_{t\in[0,A]\cup[A+G,A+G+C]} W(t) > -\varepsilon\right\} \\ &= \int_0^\infty \{\varphi_A(y-\varepsilon) - \varphi_A(y+\varepsilon)\} \int_0^\infty \varphi_G(z-y)\left\{2\Phi\left(\frac{z}{\sqrt{C}}\right) - 1\right\} dz\, dy. \end{aligned}$$

Consequently, by dominated convergence

$$r_{\mathcal{K}}(\varepsilon) \sim \kappa_{A,G,C}\varepsilon, \qquad \varepsilon \downarrow 0,$$

where

$$\kappa_{A,G,C} := \int_0^\infty \frac{2y}{A}\varphi_A(y) \int_0^\infty \varphi_G(z-y)\left\{2\Phi\left(\frac{z}{\sqrt{C}}\right) - 1\right\} dz\, dy \in (0,\infty).$$

This coefficient has an elementary closed form. Put

$$T := A + G + C, \qquad \rho := \sqrt{\frac{AC}{(A+G)(G+C)}}.$$

Then

$$\kappa_{A,G,C} = \frac{2\varphi_A(0)}{\pi} \arctan\sqrt{\frac{G}{C}} + \varphi_T(0)\left(1 + \frac{2}{\pi}\arcsin\rho\right). \tag{16}$$

Indeed, let $q(y)$ denote the inner integral in the definition of $\kappa_{A,G,C}$. Integration by parts gives

$$\kappa_{A,G,C} = 2\varphi_A(0)q(0) + 4\int_0^\infty \int_0^\infty \varphi_A(y)\varphi_G(z-y)\varphi_C(z)\, dz\, dy, \qquad q(0) = \frac{1}{\pi}\arctan\sqrt{\frac{G}{C}}.$$

After division by $\varphi_T(0)$, the product of the three densities is the joint density of a Brownian bridge on $[0,T]$ at times A and $A+G$. Its correlation is ρ. The positive-quadrant probability of a centred bivariate normal vector with correlation ρ is $1/4 + (2\pi)^{-1}\arcsin\rho$: after standardisation, representing it as $(U, \rho U + \sqrt{1-\rho^2}V)$, with independent standard normal U, V, reduces the probability to the angle of a planar sector. This proves (16). Substitution into (13) yields

$$p_{1/2}(u) \sim \frac{\kappa_{A,G,C} a_1^{3/2}}{\sqrt{2\pi}}\, u^{-2} \exp\left\{-\frac{u^2}{2a_1}\right\}, \qquad u \to \infty.$$

As $G \downarrow 0$, (16) gives $\kappa_{A,G,C} \to \sqrt{2/(\pi(A+C))}$, as expected when the gap closes. For a single interval, $\kappa = \sqrt{2/(\pi(b_1 - a_1))}$, recovering the asymptotic in [6]. Thus $H = 1/2$ is a full-contact boundary case, whereas every $H > 1/2$ lies in the dominated regime.

The first result shows that the von Mises condition (4) holds for our full-contact regime without further restrictions.

Theorem 2.2. *If $E = \mathcal{T}$ and $\sigma_*^2 > 0$, then the infimum $\mathfrak{m}_X = \inf_{t\in\mathcal{T}} X(t)$ has a strictly positive C^∞-density $f_{\mathcal{T}}$ on $\mathbb{R}$ and both*

$$h_{\mathcal{T}}(u) := \frac{f_{\mathcal{T}}(u)}{p_{\mathcal{T}}(u)} = \frac{u}{\sigma_*^2} + o(1), \qquad h'_{\mathcal{T}}(u) = \frac{1}{\sigma_*^2} + o(1), \quad u \to \infty \tag{17}$$

and

$$a_{\mathcal{T}}(u) := \frac{p_{\mathcal{T}}(u)}{f_{\mathcal{T}}(u)} = \frac{\sigma_*^2}{u} + o(u^{-2}), \qquad a'_{\mathcal{T}}(u) \sim -\frac{\sigma_*^2}{u^2}, \qquad u \to \infty. \tag{18}$$

The logarithmic asymptotic in (8) was refined for Gaussian Markov processes in [5, Theorem 3.1]. For the rough fBM considered below, [5, Theorem 3.2] already gives the lower correction bound once full support of the optimal measure is known. The matching upper bound follows from the optimal-measure density bound and an L^1 small-deviation estimate.

For the next proposition, let

$$\mathcal{T} = [a,b], \qquad 0 < a < b < \infty.$$

For $X = B_H$ with $H \in (0,1/2)$, [4, Example 2.7 and Lemma A.2] gives $S = E = [a,b]$ and $\mu_*(dt) \geq c_\mu\, dt$ on (a,b) for some $c_\mu > 0$. Our next result identifies the order of the correction to (8).

Proposition 2.3. *If $X = B_H, H \in (0,1/2)$, then there exist constants $D_\pm = D_\pm(a,b,H)$, with $0 < D_- \le D_+ < \infty$, such that for all sufficiently large u*

$$-D_+ u^{1/(1+H)} \le \log \mathbb{P}\left\{ \inf_{t\in[a,b]} X(t) > u \right\} + \frac{u^2}{2\sigma_*^2} \le -D_- u^{1/(1+H)}.$$

3. Finite Contact Sets

We consider $\mathcal{T} = K$, where K is a nonempty compact subset of $\mathbb{R}^d$, $d \ge 1$, and assume that the contact set is finite with

$$S = \{t_1, \dots, t_k\}, \qquad E = \{t \in K : m(t) = 1\} = S \cup P, \qquad P = \{s_1, \dots, s_\ell\}.$$

We further suppose that μ_* is the unique optimal probability measure. Write μ_* as in (14) and set

$$\Sigma_* = (R(t_i, t_j))_{1\le i,j\le k}.$$

The finite-support optimality relations give

$$\Sigma_* \boldsymbol{p}_* = \sigma_*^2 \mathbf{1}.$$

The matrix Σ_* is nonsingular. Indeed, if $\Sigma_* \boldsymbol{v} = 0$, then

$$\mathbf{1}^\top \boldsymbol{v} = \sigma_*^{-2} \boldsymbol{p}_*^\top \Sigma_* \boldsymbol{v} = 0,$$

and $\boldsymbol{p}_* + \varepsilon \boldsymbol{v}$ is a distinct optimal probability vector for all sufficiently small nonzero ε, unless $\boldsymbol{v} = 0$. Consequently, the regression representation (15) applies and

$$m(t) = \boldsymbol{a}(t)^\top \mathbf{1} \ge 1, \qquad t \in K,$$

where

$$\boldsymbol{a}(t) := \Sigma_*^{-1} \boldsymbol{r}(t), \qquad \boldsymbol{r}(t) := (R(t_1,t), \dots, R(t_k,t))^\top.$$

Writing

$$\boldsymbol{X}_S = (X(t_1), \dots, X(t_k))^\top$$

the decomposition (15) has a centred Gaussian residual Z, independent of $\boldsymbol{X}_S$ such that

$$\boldsymbol{a}(t_i) = \boldsymbol{e}_i, \qquad Z(t_i) = 0, \qquad i = 1, \dots, k,$$

where $\boldsymbol{e}_i$ is the i-th standard basis vector. Setting

$$\boldsymbol{S}_u := u(\boldsymbol{X}_S - u\mathbf{1})$$

yields for every $u > 0$

$$X(t) - u = u(m(t) - 1) + Z(t) + \frac{1}{u} \boldsymbol{a}(t)^\top \boldsymbol{S}_u, \qquad t \in \mathcal{T}. \tag{19}$$

Thus points outside E have a positive deterministic separation $m(t) - 1$. At a passive contact $s \in P$ by contrast $m(s) = 1$ but $Z(s)$ need not vanish, whereas $Z(t_i) = 0$ at every support contact.

3.1. Dominated contacts.

A support point t_i is called *dominated* if there exist finite positive constants β_i, C_i, λ_i and $0 < q_i < 2\beta_i$ such that for all sufficiently small admissible $h \in \mathbb{R}^d$

$$m(t_i + h) - 1 \ge \lambda_i \|h\|^{q_i}, \tag{20}$$

and for all sufficiently small admissible $h, h' \in \mathbb{R}^d$

$$\mathrm{Var}\{Z(t_i + h) - Z(t_i + h')\} \le C_i \|h - h'\|^{2\beta_i}.$$

Here admissible means that every displayed point belongs to K.

Theorem 3.1. *Under the standing assumptions of this section, if every $t_i \in S$ is dominated, then, as $u \to \infty$,*

$$\begin{aligned} \mathbb{P}\left\{ \inf_{t\in K} X(t) > u \right\} &\sim C\, \mathbb{P}\{X(t_i) > u,\ i = 1, \dots, k\} \\ &\sim \frac{C \prod_{i=1}^k \theta_i^{-1}}{(2\pi)^{k/2} \sqrt{\det \Sigma_*}}\, u^{-k} \exp\left\{ -\frac{u^2}{2\sigma_*^2} \right\}, \end{aligned}$$

where $\theta_i = p_{*,i}/\sigma_*^2$, with the convention $C = 1$ when $P = \varnothing$ and otherwise

$$C := \mathbb{P}\{Z(s) > 0,\ \forall s \in P\} > 0.$$

Remark 3.2. The regression decomposition and independence give for $s,t\in\mathcal{T}$

$$\operatorname{Var}\big(Z(t)-Z(s)\big)=\operatorname{Var}\big(X(t)-X(s)\big)-\operatorname{Var}\Big(\big(\boldsymbol{a}(t)-\boldsymbol{a}(s)\big)^{\top}\boldsymbol{X}_S\Big)\le\operatorname{Var}\big(X(t)-X(s)\big).$$

Consequently, in the definition of a dominated contact it is enough to verify the corresponding local mean-square Hölder bound for X.

Remark 3.3. Let $\mathcal{T}=[a,b]$, where $0<a<b$, and suppose that $R(a,a)>0$. If

$$R(t,a)>R(a,a),\qquad t\in(a,b],$$

then δ_a is the unique optimal measure by Corollary A.2, and

$$S=E=\{a\},\qquad m(t)=\frac{R(t,a)}{R(a,a)}.$$

If X has stationary increments, $X(0)=0$ almost surely, and $V(t):=\operatorname{Var}(X(t))$, the preceding condition is equivalent to

$$V(a+h)>V(a)+V(h),\qquad 0<h\le b-a.$$

If the preceding strict inequality holds for every $0<h\le b-a$, and for all sufficiently small $h>0$

$$V(a+h)-V(a)-V(h)\ge c_0h^q,\qquad V(h)\le Ch^{2\beta},\qquad q<2\beta,$$

then a is dominated and

$$\mathbb{P}\left\{\inf_{t\in[a,b]}X(t)>u\right\}\sim\mathbb{P}\{X(a)>u\},\qquad u\to\infty.$$

This includes fractional Brownian motion with $H>1/2$, for which $q=1$ and $\beta=H$.
The same criterion applies to

$$I(t):=\int_0^t Y(s)\,ds,$$

where Y is a centred stationary Gaussian process with continuous nonnegative covariance ρ satisfying $\rho(0)>0$, with the integral understood in mean square. The process I has stationary increments, and $\operatorname{Cov}(I(a),I(a+h)-I(a))>0$ for every $h>0$. Moreover,

$$\frac{\operatorname{Cov}(I(a),I(a+h)-I(a))}{h}\xrightarrow[h\downarrow0]{}\int_0^a\rho(v)\,dv>0,\qquad\operatorname{Var}(I(t)-I(s))\le\rho(0)|t-s|^2.$$

The Gaussian moment bounds associated with this increment estimate give a continuous version of I by the Kolmogorov–Chentsov theorem; we use this version throughout. Thus $q=\beta=1$. The covariance criterion can also be verified directly for the Riemann–Liouville process

$$X_H^{\mathrm{RL}}(t):=\int_0^t(t-s)^{H-\frac12}\,dB(s),\qquad H\in(1/2,1).$$

Its covariance is strictly increasing in $t\ge a$ with the first argument fixed at a. Differentiating the covariance integral and applying Itô's isometry give

$$m(a+h)-1\sim\frac{H}{a}h,\quad h\downarrow0,\qquad\operatorname{Var}\{X_H^{\mathrm{RL}}(t)-X_H^{\mathrm{RL}}(s)\}\le C_H|t-s|^{2H}$$

for some $C_H<\infty$. For the increment bound, rescaling the integrals in Itô's isometry uses

$$\int_0^\infty\big((v+1)^{H-1/2}-v^{H-1/2}\big)^2\,dv<\infty.$$

Hence $q=1$ and $\beta=H$.

Let

$$I_H(t):=\int_0^t B_H(r)\,dr,\qquad H\in(0,1),$$

where B_H is a standard fractional Brownian motion, and fix $0<a<b<\infty$. Write

$$R_H(r,s):=\operatorname{Cov}(B_H(r),B_H(s)).$$

For $t>a$ we have

$$\operatorname{Cov}(I_H(a),I_H(t))-\operatorname{Var}(I_H(a))=\int_0^a\int_a^t R_H(r,s)\,ds\,dr>0.$$

Thus δ_a is the unique optimal measure and

$$S=E=\{a\},\qquad P=\varnothing.$$

Moreover we have

$$\sigma_*^2 = \frac{a^{2H+2}}{2H+2}, \qquad m(a+h) - 1 \sim \frac{H+1}{a}h, \qquad h \downarrow 0,$$

while for $s, t \in [a, b]$

$$\mathrm{Var}(I_H(t) - I_H(s)) \le C|t-s|^2.$$

Hence $q = \beta = 1$, so a is dominated and

$$\mathbb{P}\left\{\inf_{t\in[a,b]} I_H(t) > u\right\} \sim \mathbb{P}\{I_H(a) > u\} \sim \frac{a^{H+1}}{2\sqrt{\pi(H+1)}\,u} \exp\left\{-\frac{H+1}{a^{2H+2}}u^2\right\}, \qquad u \to \infty.$$

Let X be the centred Gaussian process with covariance

$$R(s,t) = \frac{2st}{s+t}, \qquad s, t > 0;$$

see [9]. On $\mathcal{T} = [a, b]$, where $0 < a < b < \infty$

$$m(t) = \frac{R(t,a)}{R(a,a)} = \frac{2t}{a+t} > 1, \qquad t > a.$$

Thus δ_a is the unique optimal measure and

$$\mu_* = \delta_a, \qquad S = E = \{a\}, \qquad \sigma_*^2 = a.$$

For the regression residual $Z(t) = X(t) - m(t)X(a)$

$$m(a+h) - 1 = \frac{h}{2a+h} \ge \frac{h}{2b}, \qquad \mathrm{Var}\{Z(t) - Z(s)\} \le \frac{(t-s)^2}{t+s} \le \frac{(t-s)^2}{2a}.$$

Hence $q = \beta = 1$ and

$$\mathbb{P}\left\{\inf_{t\in[a,b]} X(t) > u\right\} \sim \mathbb{P}\{X(a) > u\}, \qquad u \to \infty.$$

Notice that $\lim_{t\to\infty} R(s,t) = 2s > 0$, so this process does not satisfy the additional condition in [1, Proposition 3.1].

Let X be the fractional Brownian sheet with common Hurst index $H \in (1/2, 1)$ and covariance

$$R(s,t) = \prod_{j=1}^{2} \frac{s_j^{2H} + t_j^{2H} - |s_j - t_j|^{2H}}{2}.$$

On $\mathcal{T} = [a, b]^2$, where $0 < a < b < \infty$, put $t_0 = (a, a)$. The normalised potential of δ_{t_0} is

$$m(t) = m_H(t_1) m_H(t_2), \qquad m_H(x) := \frac{x^{2H} + a^{2H} - (x-a)^{2H}}{2a^{2H}}.$$

Strict superadditivity of $x \mapsto x^{2H}$ gives $m_H(x) > 1$ for $x > a$. Therefore we obtain

$$\mu_* = \delta_{t_0}, \qquad S = E = \{t_0\}, \qquad \sigma_*^2 = a^{4H}.$$

Moreover we have

$$m_H(a+x) - 1 = \frac{H}{a}x + o(x), \qquad x \downarrow 0$$

and uniformly on $[a, b]^2$

$$\mathrm{Var}\{X(t) - X(s)\} \le C\|t-s\|^{2H}.$$

The same bound holds for the regression residual. For sufficiently small $h = (h_1, h_2)$ with $h_1, h_2 \ge 0$

$$m(t_0 + h) - 1 \ge \max_{j=1,2}\{m_H(a + h_j) - 1\} \ge c \max(h_1, h_2) \ge \frac{c}{\sqrt{2}}\|h\|$$

for some $c > 0$. Thus $q = 1$, $\beta = H$ and $q < 2\beta$. An application of Theorem 3.1 gives

$$\mathbb{P}\left\{\inf_{t\in[a,b]^2} X(t) > u\right\} \sim \mathbb{P}\{X(a,a) > u\}, \qquad u \to \infty.$$

Example 3.4. Let ξ_1, ξ_2, ζ be independent standard normal rvs, let $\mathcal{T} = [-1, 1]$, and put

$$g(t) := t^2(1-t^2), \qquad a_1(t) := \frac{1-t+g(t)}{2}, \qquad a_2(t) := \frac{1+t+g(t)}{2}.$$

Define

$$X(t) := a_1(t)\xi_1 + a_2(t)\xi_2 + (1-t^2)\zeta, \qquad t \in [-1, 1].$$

For

$$\mu_* = \frac{1}{2}(\delta_{-1} + \delta_1)$$

we have

$$X_{\mu_*} = \frac{\xi_1 + \xi_2}{2}, \qquad \sigma_*^2 = \frac{1}{2}, \qquad m(t) = a_1(t) + a_2(t) = 1 + g(t).$$

Hence μ_* is optimal and

$$S = \{-1, 1\}, \qquad E = \{-1, 0, 1\}, \qquad P = \{0\}.$$

Any other optimal measure must be supported on E. If its masses at $-1, 0, 1$ are x, y, z, respectively, then

$$\mathrm{Var}\left(\int_{\mathcal{T}} X\, d\mu\right) = (x + y/2)^2 + (z + y/2)^2 + y^2 \geq \frac{1}{2} + y^2.$$

Thus equality is possible only when $y = 0$ and $x = z = 1/2$, proving uniqueness of μ_*.
The support-regression residual is $Z(t) = (1-t^2)\zeta$. Since

$$g(-1+h) = 2h + O(h^2), \qquad g(1-h) = 2h + O(h^2),$$

and

$$\mathrm{Var}\{Z(t) - Z(s)\} = (t^2 - s^2)^2 \leq 4|t-s|^2,$$

both support contacts are dominated with $q = \beta = 1$. At the passive contact, $Z(0) = \zeta$, so

$$C = \mathbb{P}\{Z(0) > 0\} = \frac{1}{2}.$$

Consequently, we have

$$\mathbb{P}\left\{\inf_{t\in[-1,1]} X(t) > u\right\} \sim \frac{1}{2}\,\mathbb{P}\{\xi_1 > u,\ \xi_2 > u\}, \qquad u \to \infty.$$

3.2. **Critical rough and smooth contacts.** Throughout this subsection, $\mathcal{T} = [a, b]$, where $-\infty < a < b < \infty$. For simplicity, we consider only the case where the optimal measure is a Dirac measure at a single point b, with $R(b, b) > 0$, and

$$E = S = \{b\}.$$

Then $\mu_* = \delta_b$ is the unique optimal measure and

$$m(t) = \frac{R(t, b)}{R(b, b)}, \qquad m(b) = 1, \qquad m(t) > 1, \quad t \in \mathcal{T} \setminus \{b\}.$$

Since the support is a singleton, there is no distinction between the optimal-measure residual and the support-regression residual. Thus, throughout this subsection

$$Z(t) := X(t) - m(t)X(b), \qquad t \in \mathcal{T},$$

is independent of $X(b)$, satisfies $Z(b) = 0$, and has continuous sample paths.
We say that b is a tangent-critical endpoint contact with exponent $\beta \in (0, 1]$, coefficient $\lambda > 0$, and tangent process W, if the following conditions hold. First, for some $C > 0$ and $0 < r_0 \leq b - a$

$$\mathrm{Var}\{Z(b-h) - Z(b-h')\} \leq C|h - h'|^{2\beta}, \qquad 0 \leq h, h' \leq r_0. \tag{21}$$

Second, as $h \downarrow 0$

$$m(b-h) - 1 = \lambda h^{2\beta} + o(h^{2\beta}). \tag{22}$$

Third, for every $L > 0$, as $r \downarrow 0$, the following convergence of finite-dimensional distributions holds:

$$\left\{r^{-\beta} Z(b - rx),\ 0 \leq x \leq L\right\} \Rightarrow \{W(x),\ 0 \leq x \leq L\}, \tag{23}$$

where $W = \{W(x),\, x \geq 0\}$ is a centred GP with continuous sample paths and $W(0) = 0$. When $\beta \in (0, 1)$, we call the contact rough-critical.
A smooth-critical endpoint contact is the differentiable tangent-critical case $\beta = 1$, in which the residual has a linear L^2-tangent. More precisely, we assume that as $h \downarrow 0$

$$m(b-h) - 1 = \lambda h^2 + o(h^2)$$

for some $\lambda > 0$, and that for some centred Gaussian rv η

$$\frac{Z(b-h)}{h} \longrightarrow \eta \quad \text{in } L^2, \qquad h \downarrow 0.$$

We also assume that (21) holds for some $C > 0, 0 < r_0 \le b - a$ with $\beta = 1$.

Theorem 3.5. *Under the standing assumptions, let $\mathcal{T} = [a, b]$ with $a < b$. Assume that $R(b, b) > 0$, that $\mu_* = \delta_b$ is optimal, and that $E = \{b\}$. If b is a tangent-critical endpoint contact, then $M := \inf_{x\ge 0}\{W(x) + \lambda x^{2\beta}\} \in (-\infty, 0]$ almost surely, and*

$$p_{\mathcal{T}}(u) \sim C_M \, \mathbb{P}\{X(b) > u\}, \qquad u \to \infty,$$

where

$$C_M := \mathbb{E}\left\{\exp\left\{\frac{M}{R(b,b)}\right\}\right\} \in (0, 1].$$

Corollary 3.6. *Under the standing assumptions assume that $R(b, b) > 0$, that $\mu_* = \delta_b$ is optimal, and that $E = \{b\}$. If b is a smooth-critical endpoint contact and $v^2 := \mathrm{Var}(\eta)$, then*

$$p_{\mathcal{T}}(u) \sim \left\{\frac{1}{2} + \frac{1}{2}\left(1 + \frac{v^2}{2\lambda R(b,b)}\right)^{-1/2}\right\} \mathbb{P}\{X(b) > u\}, \qquad u \to \infty.$$

Remark 3.7. The left-endpoint versions follow by applying the results to $\widetilde{X}(t) := X(a + b - t)$.
For an interior contact t_0 with $E = S = \{t_0\}$, use $m(t) = R(t, t_0)/R(t_0, t_0)$ and $Z(t) = X(t) - m(t)X(t_0)$. The local increment bound is assumed to hold across both sides. Suppose also that for every $L > 0$

$$\sup_{|x|\le L} \left|r^{-2\beta}\{m(t_0 + rx) - 1\} - d(x)\right| \longrightarrow 0, \qquad r \downarrow 0,$$

and

$$\{r^{-\beta} Z(t_0 + rx),\ |x| \le L\} \Rightarrow \{W(x),\ |x| \le L\} \quad \text{in } C[-L, L], \qquad r \downarrow 0,$$

and that $m(t_0 + h) - 1 \ge c|h|^{2\beta}$ for some $c > 0$ and all sufficiently small $|h|$. Then the conclusion of Theorem 3.5 holds with b replaced by t_0 and with

$$M = \inf_{x\in\mathbb{R}}\{W(x) + d(x)\} \in (-\infty, 0] \quad \text{almost surely.}$$

In the symmetric smooth case, $W(x) = \eta x$ and $d(x) = \lambda x^2$, so $M = -\eta^2/(4\lambda)$. In the asymmetric case

$$M = \min\{M_-, M_+\}, \qquad M_- := \inf_{x\le 0}\{W(x) + d(x)\}, \qquad M_+ := \inf_{x\ge 0}\{W(x) + d(x)\},$$

where the two branches come from the joint two-sided tangent.
To justify the extension, put $r_u = u^{-1/\beta}$ and apply Lemma 4.4 separately to $x \mapsto uZ(t_0 \pm r_u x)$, with drifts $u^2\{m(t_0 \pm r_u x) - 1\}$, on $[0, r_0/r_u]$ for a sufficiently small $r_0 > 0$. The increment bounds are uniform in u, and both drifts are bounded below by $cx^{2\beta}$. The probability that either tail beyond L affects the local minimum is therefore $O(L^{-\beta})$, uniformly in u. The same lemma applied to the two limiting branches proves that $M > -\infty$; here $d(x) \ge c|x|^{2\beta}$ follows from the assumed convergence, and $W(0) = d(0) = 0$. Joint convergence on $[-L, L]$ gives convergence of the compact-window random threshold. Since $E = \{t_0\}$, compactness also excludes the region away from t_0. The independent one-point overshoot argument used in the proof of Theorem 3.5 then gives the factor $\mathbb{E}\exp\{M/R(t_0, t_0)\}$.

Example 3.8. Let B be a standard Brownian motion independent of $\xi \sim N(0, 1)$, and define

$$X(t) := \xi(1 + \lambda t) + \sigma B(t), \qquad t \in [0, 1], \qquad \lambda, \sigma > 0.$$

Then

$$\mu_* = \delta_0, \qquad S = E = \{0\}, \qquad m(t) = 1 + \lambda t, \qquad Z(t) = \sigma B(t).$$

Brownian scaling implies the following equality in distribution for every $L > 0$ and all sufficiently small $r > 0$

$$\{r^{-1/2} Z(rx),\ 0 \le x \le L\} \overset{d}{=} \{\sigma B(x),\ 0 \le x \le L\}.$$

Thus 0 is a rough-critical left-endpoint contact with $\beta = 1/2$, and

$$M = \inf_{x\ge 0}\{\sigma B(x) + \lambda x\}.$$

Since $-M$ is exponential with rate $2\lambda/\sigma^2$, the left-endpoint version of Theorem 3.5 gives

$$\mathbb{P}\left\{\inf_{0\le t\le 1} X(t) > u\right\} \sim \frac{2\lambda}{2\lambda + \sigma^2}\, \mathbb{P}\{\xi > u\}, \qquad u \to \infty.$$

Example 3.9. Let B be a standard Brownian motion. Fix $0 < c < 1$, $\alpha > 1$ and set

$$X(t) := \frac{B(t)}{t^\alpha}, \qquad t \in \mathcal{T} = [c, 1].$$

Since $X(1) = B(1)$ and

$$m(t) = \frac{\mathrm{Cov}(X(t), X(1))}{\mathrm{Var}\, X(1)} = t^{1-\alpha},$$

the optimality criterion gives

$$\mu_* = \delta_1, \qquad E = S = \{1\}, \qquad Z(t) = t^{-\alpha}\{B(t) - tB(1)\}.$$

As $h \downarrow 0$

$$m(1-h) - 1 = (\alpha - 1)h + o(h),$$

locally uniformly after replacing h by rx with x in a compact interval. Moreover, for $s \le t$ near 1

$$\mathrm{Var}(X(t) - X(s)) = \frac{t-s}{t^{2\alpha}} + s\big(t^{-\alpha} - s^{-\alpha}\big)^2 \le C|t-s|.$$

The same bound holds for the regression residual Z.
For fixed $L < \infty$ and sufficiently small r, set

$$\widetilde{B}_r(x) := r^{-1/2}\{B(1) - B(1 - rx)\}, \qquad 0 \le x \le L.$$

Then $\widetilde{B}_r$ is a standard Brownian motion on $[0, L]$, and

$$r^{-1/2} Z(1 - rx) = (1 - rx)^{-\alpha}\{-\widetilde{B}_r(x) + r^{1/2} x B(1)\} \Rightarrow -\widetilde{B}(x)$$

in $C[0, L]$, where $\widetilde{B}$ is a standard Brownian motion. Hence 1 is rough-critical with

$$\beta = \frac{1}{2}, \qquad \lambda = \alpha - 1.$$

By Brownian symmetry, its variational rv may be written as

$$M = \inf_{x \ge 0}\{\widetilde{B}(x) + (\alpha - 1)x\}.$$

Since $-M$ is exponential with rate $2(\alpha - 1)$, Theorem 3.5 yields

$$\mathbb{P}\left\{\inf_{t \in [c,1]} \frac{B(t)}{t^\alpha} > u\right\} \sim \frac{2(\alpha - 1)}{2\alpha - 1}\, \mathbb{P}\{B(1) > u\}, \qquad u \to \infty.$$

Remark 3.10. Let $Y \sim N(0, \tau^2)$, $\tau > 0$, and let V be a centred Gaussian random element of $C^1[0, 1]$, independent of Y, with $V(1) = 0$ almost surely. Suppose that $m \in C[0, 1]$ satisfies

$$m(1) = 1, \qquad m(t) > 1, \quad 0 \le t < 1, \qquad m(1 - h) = 1 + \lambda h^2 + o(h^2)$$

for some $\lambda > 0$, and put $X(t) = m(t)Y + V(t)$. Then

$$\mu_* = \delta_1, \qquad E = S = \{1\}, \qquad \sigma_*^2 = \tau^2, \qquad Z = V.$$

Moreover, Fernique's theorem and the fundamental theorem of calculus give

$$\mathbb{E}\{(V(t) - V(s))^2\} \le |t - s|^2 \mathbb{E}\{\|V'\|_\infty^2\}, \qquad \frac{V(1-h)}{h} \longrightarrow -V'(1) \quad \text{in } L^2, \qquad h \downarrow 0.$$

Thus Corollary 3.6 applies with $v^2 = \mathrm{Var}(V'(1))$ and yields

$$\mathbb{P}\left\{\inf_{0 \le t \le 1} X(t) > u\right\} \sim \left\{\frac{1}{2} + \frac{1}{2}\left(1 + \frac{v^2}{2\lambda\tau^2}\right)^{-1/2}\right\} \mathbb{P}\{Y > u\}, \qquad u \to \infty.$$

4. Proofs

Standing notation for finite-contact proofs: Whenever a finite-support optimal measure is under consideration, we write

$$A_u := \{X(t_i) > u,\ i = 1, \dots, k\}, \qquad B_u := \left\{\inf_{t\in\mathcal{T}} X(t) > u\right\},$$

and

$$\mathbb{P}_u(\cdot) := \mathbb{P}(\cdot \mid A_u), \qquad \boldsymbol{S}_u := u(\boldsymbol{X}_S - u\mathbf{1}).$$

If the passive set is finite, we write

$$P = \{s_1, \dots, s_\ell\}.$$

Since $B_u \subset A_u$, all exact-asymptotic proofs below reduce to proving

$$\lim_{u\to\infty} \mathbb{P}_u(B_u) = C$$

for the appropriate constant C. The residual process Z is independent of $\boldsymbol{X}_S$ and remains so after conditioning on A_u. Setting

$$\boldsymbol{\theta} := \Sigma_*^{-1}\mathbf{1} = \frac{\boldsymbol{p}_*}{\sigma_*^2}$$

we have $\theta_i > 0$ for every i. For $\boldsymbol{x} \in (0,\infty)^k$ a change of variables in the Gaussian density shows that the conditional density of $\boldsymbol{S}_u$ under $\mathbb{P}_u$ is

$$f_u(\boldsymbol{x}) = \frac{g_u(\boldsymbol{x})}{\displaystyle\int_{(0,\infty)^k} g_u(\boldsymbol{y})\,d\boldsymbol{y}}, \qquad g_u(\boldsymbol{x}) := \exp\left\{-\boldsymbol{\theta}^\top\boldsymbol{x} - \frac{1}{2u^2}\boldsymbol{x}^\top\Sigma_*^{-1}\boldsymbol{x}\right\}.$$

Since Σ_*^{-1} is positive definite

$$0 \le g_u(\boldsymbol{x}) \le \exp\{-\boldsymbol{\theta}^\top\boldsymbol{x}\},$$

and the function on the right is integrable on $(0,\infty)^k$. Dominated convergence therefore gives under $\mathbb{P}_u$

$$\boldsymbol{S}_u \Rightarrow \boldsymbol{\mathcal{E}} := (\mathcal{E}_1, \dots, \mathcal{E}_k), \qquad u \to \infty, \tag{24}$$

where the coordinates of $\boldsymbol{\mathcal{E}}$ are independent exponential random variables with respective rates $\theta_1, \dots, \theta_k$. In particular, $\boldsymbol{S}_u$ is tight under $\mathbb{P}_u$.

In the one-point endpoint proofs below we take $t_1 = b$. Then $A_u = \{X(b) > u\}$, $S_u = u(X(b) - u)$, and the preceding convergence holds with exponential rate

$$\theta = \frac{1}{R(b,b)}.$$

4.1. Proofs for full contact.

Proof of Theorem 2.2. Set

$$V := -\inf_{t\in\mathcal{T}} Z_*(t), \qquad r(\varepsilon) := \mathbb{P}\{V < \varepsilon\}, \qquad \varepsilon > 0,$$

and write $s := \sigma_*^2$. Path continuity and compactness of $\mathcal{T}$ make V finite almost surely, while (10) gives $V \ge 0$. Since $E = \mathcal{T}$

$$\mathfrak{m}_X = X_{\mu_*} - V,$$

where $X_{\mu_*} \sim N(0, s)$ is independent of V. Consequently, we have

$$f_{\mathcal{T}}(u) = \frac{1}{\sqrt{2\pi s}}\mathbb{E}\exp\left\{-\frac{(u+V)^2}{2s}\right\}.$$

This density is strictly positive. Every derivative of the Gaussian density is bounded, so differentiation under the expectation also shows that $f_{\mathcal{T}} \in C^\infty(\mathbb{R})$. With the notation

$$I(u) := \int_0^\infty \exp\left\{-\frac{u\varepsilon}{s} - \frac{\varepsilon^2}{2s}\right\} r(\varepsilon)\,d\varepsilon$$

in view of (11)

$$p_{\mathcal{T}}(u) = \frac{e^{-u^2/(2s)}}{\sqrt{2\pi s}} I(u). \tag{25}$$

Since zero belongs to the support of the centred Gaussian law of Z_* on $C(\mathcal{T})$, we have $r(\delta) > 0$ for every $\delta > 0$. Define the finite measure

$$\nu(d\varepsilon) := e^{-\varepsilon^2/(2s)} r(\varepsilon)\, d\varepsilon, \qquad \varepsilon > 0,$$

and the probability measure

$$\Pi_u(d\varepsilon) := \frac{e^{-u\varepsilon/s}\nu(d\varepsilon)}{I(u)}.$$

For every fixed $\delta > 0$, we have $\nu((0,\delta/2)) > 0$ and

$$I(u) \geq e^{-u\delta/(2s)}\nu((0,\delta/2)), \qquad u > 0.$$

Since $0 \leq r \leq 1$, the second moment of ν is finite. Splitting at δ, we therefore obtain

$$\begin{aligned}\int_0^\infty \varepsilon^2\, \Pi_u(d\varepsilon) &\leq \delta^2 + \frac{e^{-u\delta/s}\int_0^\infty \varepsilon^2\, \nu(d\varepsilon)}{e^{-u\delta/(2s)}\nu((0,\delta/2))} \\ &\leq \delta^2 + C_\delta e^{-u\delta/(2s)}.\end{aligned}$$

Letting first $u \to \infty$ and then $\delta \downarrow 0$ gives

$$\lim_{u\to\infty} \int_0^\infty \varepsilon^2\, \Pi_u(d\varepsilon) = 0. \tag{26}$$

Write $\bar{\varepsilon}_u := \mathbb{E}_{\Pi_u}\varepsilon$ and $\tau_u^2 := \mathrm{Var}_{\Pi_u}(\varepsilon)$. By (26), both tend to zero. Differentiation under the integral sign is justified by the Gaussian factor and $0 \leq r \leq 1$, and gives

$$(\log I)'(u) = -\frac{\bar{\varepsilon}_u}{s}, \qquad (\log I)''(u) = \frac{\tau_u^2}{s^2}.$$

Since $h_{\mathcal{T}} = -(\log p_{\mathcal{T}})'$, (25) therefore yields

$$h_{\mathcal{T}}(u) = \frac{u + \bar{\varepsilon}_u}{s}, \qquad h'_{\mathcal{T}}(u) = \frac{1}{s} - \frac{\tau_u^2}{s^2}.$$

These identities prove (17). Finally,

$$a_{\mathcal{T}}(u) = \frac{s}{u + \bar{\varepsilon}_u} = \frac{s}{u} + o(u^{-2}), \qquad a'_{\mathcal{T}}(u) = -\frac{s - \tau_u^2}{(u+\bar{\varepsilon}_u)^2} \sim -\frac{s}{u^2},$$

which proves (18). □

Lemma 4.1. *Assume that $E = \mathcal{T}$ and $\sigma_*^2 > 0$ and set*

$$\mathfrak{m}_{Z_*} := \inf_{t\in\mathcal{T}} Z_*(t), \qquad r(\varepsilon) := \mathbb{P}\{\mathfrak{m}_{Z_*} > -\varepsilon\}.$$

If for some $\beta, A_1, A_2, \varepsilon_0 > 0$

$$\exp\{-A_2\varepsilon^{-1/\beta}\} \leq \mathbb{P}\{\mathfrak{m}_{Z_*} > -\varepsilon\} \leq \exp\{-A_1\varepsilon^{-1/\beta}\}, \qquad 0 < \varepsilon < \varepsilon_0, \tag{27}$$

then there exist constants $0 < C_- \leq C_+ < \infty$ such that for all sufficiently large u

$$-C_+ u^{1/(\beta+1)} \leq \log p_{\mathcal{T}}(u) + \frac{u^2}{2\sigma_*^2} \leq -C_- u^{1/(\beta+1)}. \tag{28}$$

Proof of Lemma 4.1. Write

$$I_u := \int_0^\infty \exp\left\{-\frac{u\varepsilon}{\sigma_*^2} - \frac{\varepsilon^2}{2\sigma_*^2}\right\} r(\varepsilon)\, d\varepsilon$$

and put $\kappa := 1/(\beta+1)$ and $\varepsilon_u := u^{-\beta/(\beta+1)}$. For all large u, $2\varepsilon_u < \varepsilon_0$. Splitting at ε_u, using monotonicity of r on the first part and $r \leq 1$ on the second, gives

$$I_u \leq \frac{\sigma_*^2}{u}\left[r(\varepsilon_u) + e^{-u\varepsilon_u/\sigma_*^2}\right] \leq \frac{\sigma_*^2}{u}\left[e^{-A_1 u^\kappa} + e^{-u^\kappa/\sigma_*^2}\right].$$

Hence $\log I_u \leq -c_2 u^\kappa$ for some $c_2 > 0$ and all large u.
For the lower bound, monotonicity and (27) give $r(\varepsilon) \geq r(\varepsilon_u) \geq e^{-A_2 u^\kappa}$ on $[\varepsilon_u, 2\varepsilon_u]$. Consequently, we obtain

$$I_u \geq \varepsilon_u \exp\left\{-\left(A_2 + \frac{2}{\sigma_*^2}\right) u^\kappa - 1\right\}$$

for all sufficiently large u; here the final constant bounds $\varepsilon^2/(2\sigma_*^2)$ uniformly on $[\varepsilon_u, 2\varepsilon_u]$. Since $|\log \varepsilon_u| = O(\log u) = o(u^\kappa)$, we obtain $\log I_u \geq -c_3 u^\kappa$. Finally, (11) gives

$$\log p_{\mathcal{T}}(u) + \frac{u^2}{2\sigma_*^2} = \log I_u - \log(\sqrt{2\pi}\sigma_*),$$

and the claimed two-sided bound follows after adjusting the constants. □

Proof of Proposition 2.3. Let μ_* be an optimal measure. By [4, Example 2.7 and Lemma A.2], there exists $c_\mu > 0$ such that

$$\int_a^b R(s,t)\,\mu_*(ds) = \sigma_*^2, \qquad t \in [a,b], \qquad \mu_*(dt) \geq c_\mu\,dt \quad \text{on } (a,b).$$

In particular, $\sigma_*^2 > 0$ and $E = [a,b]$. Define

$$Y := \int_a^b X(t)\,\mu_*(dt), \qquad Z_*(t) := X(t) - Y, \qquad \mathfrak{m}_{Z_*} := \inf_{t\in[a,b]} Z_*(t).$$

Then $Y \sim N(0,\sigma_*^2)$ is independent of Z_*, and

$$\int_a^b Z_*(t)\,\mu_*(dt) = 0 \qquad \text{a.s.}$$

We first record the small-deviation estimates needed below. For some $c, C, \varepsilon_0 > 0$, and all $0 < \varepsilon < \varepsilon_0$

$$\begin{aligned} &\mathbb{P}\left\{\sup_{0\leq t\leq b}|X(t)| \leq \varepsilon\right\} \geq \exp\{-C\varepsilon^{-1/H}\}, \\ &\mathbb{P}\left\{\int_a^b |X(t)|\,dt \leq \varepsilon\right\} \leq \exp\{-c\varepsilon^{-1/H}\}. \end{aligned} \tag{29}$$

The first inequality is the sup-norm small-deviation estimate of [10]; the second follows from [11, Theorem 5.7 and equation (5.13), with $N = D = q = 1$]. For the latter application, normalised Lebesgue measure on $[a,b]$ is the self-similar measure generated by the two half-interval similarities; the open set condition holds with open set (a,b). The result therefore applies directly to this interval, also when $a > 0$, and normalisation only changes the constants.

Put

$$r(\varepsilon) := \mathbb{P}\{\mathfrak{m}_{Z_*} > -\varepsilon\}, \qquad \|f\|_1 := \int_a^b |f(t)|\,dt.$$

If $\sup_{0\leq t\leq b}|X(t)| \leq \varepsilon/3$, then for $t \in [a,b]$

$$|Z_*(t)| = \left|\int_a^b \{X(t) - X(s)\}\,\mu_*(ds)\right| \leq \frac{2\varepsilon}{3}.$$

The first estimate in (29) therefore gives for some $0 < A_2 < \infty$ and all sufficiently small $\varepsilon > 0$

$$r(\varepsilon) \geq \mathbb{P}\left\{\sup_{0\leq t\leq b}|X(t)| \leq \frac{\varepsilon}{3}\right\} \geq \exp\{-A_2\varepsilon^{-1/H}\}.$$

For the upper bound on r, the anchoring identity implies that on $\{\mathfrak{m}_{Z_*} > -\varepsilon\}$

$$c_\mu\|Z_*\|_1 \leq \int_a^b |Z_*(t)|\,\mu_*(dt) = 2\int_a^b (Z_*(t))_-\,\mu_*(dt) \leq 2\varepsilon.$$

Put $K := 2/c_\mu + b - a$. Intersecting this event with $\{|Y| \leq \varepsilon\}$, and using $X = Y + Z_*$ and independence, gives

$$r(\varepsilon)\,\mathbb{P}\{|Y| \leq \varepsilon\} \leq \mathbb{P}\{\|X\|_1 \leq K\varepsilon\}.$$

Since $\mathbb{P}\{|Y| \leq \varepsilon\} \asymp \varepsilon$, the second estimate in (29) yields

$$r(\varepsilon) \leq C_1\varepsilon^{-1}\exp\{-c_1\varepsilon^{-1/H}\} \leq \exp\{-A_1\varepsilon^{-1/H}\}$$

for some $A_1 > 0$ and all sufficiently small $\varepsilon > 0$. Thus both inequalities in (27) hold with $\beta = H$, and Lemma 4.1 proves the proposition. □

4.2. **Proofs for finite contact.**

Lemma 4.2. *Assume that*

$$\mu_* = \sum_{i=1}^{k} p_{*,i}\delta_{t_i}, \qquad p_{*,i} > 0, \quad i = 1, \dots, k,$$

is the unique optimal probability measure. If $P = E \setminus S = \{s_1, \dots, s_\ell\}$ is nonempty, then

$$\mathbb{P}\{Z(s_j) \geq 0,\ j = 1, \dots, \ell\} = \mathbb{P}\{Z(s_j) > 0,\ j = 1, \dots, \ell\} > 0.$$

Proof. Define

$$\boldsymbol{\Delta} := (Z(s_1), \dots, Z(s_\ell))^\top$$

and let $L \subseteq \mathbb{R}^\ell$ be the linear support of its centred Gaussian law. Suppose that

$$\mathbb{P}\{\Delta_j > 0,\ j = 1, \dots, \ell\} = 0.$$

Since the Gaussian law has full support on L, $L \cap (0, \infty)^\ell = \varnothing$. Gordan's alternative provides a nonzero vector $\boldsymbol{\lambda} \in [0, \infty)^\ell$ such that

$$\boldsymbol{\lambda}^\top \boldsymbol{x} = 0, \qquad \boldsymbol{x} \in L.$$

Consequently, $\boldsymbol{\lambda}^\top \boldsymbol{\Delta} = 0$ almost surely.
Let

$$\boldsymbol{A} := (a_i(s_j))_{\substack{1 \leq i \leq k, \\ 1 \leq j \leq \ell}}, \qquad \boldsymbol{c} := \boldsymbol{A}\boldsymbol{\lambda}.$$

Since $m(s_j) = \sum_{i=1}^{k} a_i(s_j) = 1$

$$\sum_{i=1}^{k} c_i = \sum_{j=1}^{\ell} \lambda_j m(s_j) = \sum_{j=1}^{\ell} \lambda_j.$$

As every $p_{*,i} > 0$, for all sufficiently small $\varepsilon > 0$

$$\mu_\varepsilon := \sum_{i=1}^{k} (p_{*,i} - \varepsilon c_i)\delta_{t_i} + \varepsilon \sum_{j=1}^{\ell} \lambda_j \delta_{s_j}$$

is a probability measure. Indeed, its coefficients are nonnegative and the preceding identity shows that their sum is one. Writing

$$\boldsymbol{X}_P := (X(s_1), \dots, X(s_\ell))^\top,$$

we have $\boldsymbol{X}_P = \boldsymbol{A}^\top \boldsymbol{X}_S + \boldsymbol{\Delta}$, and hence

$$\begin{aligned} \int_{\mathcal{T}} X(t)\, \mu_\varepsilon(dt) - \int_{\mathcal{T}} X(t)\, \mu_*(dt) &= \varepsilon \left(\boldsymbol{\lambda}^\top \boldsymbol{X}_P - \boldsymbol{c}^\top \boldsymbol{X}_S \right) \\ &= \varepsilon \boldsymbol{\lambda}^\top \left(\boldsymbol{X}_P - \boldsymbol{A}^\top \boldsymbol{X}_S \right) \\ &= \varepsilon \boldsymbol{\lambda}^\top \boldsymbol{\Delta} = 0 \qquad \text{a.s.} \end{aligned}$$

Thus μ_ε is another optimal measure. It is distinct from μ_*, because $\boldsymbol{\lambda} \neq 0$ and $s_j \notin S$, contradicting uniqueness. Therefore

$$\mathbb{P}\{\Delta_j > 0,\ j = 1, \dots, \ell\} > 0.$$

In particular, no Δ_j is identically zero. Each coordinate is therefore a nondegenerate Gaussian rv, so

$$\mathbb{P}\{\Delta_j = 0\} = 0, \qquad j = 1, \dots, \ell.$$

Consequently, we obtain

$$\mathbb{P}\{\Delta_j \geq 0,\ j = 1, \dots, \ell\} = \mathbb{P}\{\Delta_j > 0,\ j = 1, \dots, \ell\} > 0,$$

as claimed. □

Proof of Theorem 3.1. Lemma 4.2 yields

$$C := \mathbb{P}\{Z(s) > 0,\ s \in P\} > 0$$

and almost surely no passive value is zero. When $P = \varnothing$, these conditions are void and $C = 1$.
For each i, choose an even integer $N_i \geq 2$ and an exponent γ_i such that

$$N_i \left(\beta_i - \frac{q_i}{2} \right) > d, \qquad \frac{q_i}{2} < \gamma_i < \min \left\{ q_i, \beta_i - \frac{d}{N_i} \right\}.$$

On sufficiently small relative neighbourhoods U_i of t_i, the dominated-contact assumptions and Gaussian moment estimates give

$$m(t_i+h)-1 \geq \lambda_i \|h\|^{q_i}, \qquad \mathbb{E}\left\{|Z(t_i+h)-Z(t_i+h')|^{N_i}\right\} \leq C_{i,N_i}\|h-h'\|^{N_i\beta_i}.$$

Since every compact subset of $\mathbb{R}^d$ has Euclidean covering numbers bounded by $C\varepsilon^{-d}$, the dyadic-net proof of the Kolmogorov–Chentsov theorem applies locally on K. After shrinking U_i, it therefore gives an almost surely finite rv K_i such that

$$|Z(t_i+h)| \leq K_i\|h\|^{\gamma_i}, \qquad t_i+h \in U_i,$$

using $Z(t_i)=0$ and the given continuous version.
For $\boldsymbol{x} \in (0,\infty)^k$ and $z \in C(K)$, put

$$H_u(\boldsymbol{x},z) := \mathbf{1}\left\{\inf_{t\in K}\left[u\{m(t)-1\}+z(t)+\frac{\boldsymbol{a}(t)^\top \boldsymbol{x}}{u}\right] > 0\right\}.$$

We claim that, for every fixed $\boldsymbol{x} \in (0,\infty)^k$ as $u \to \infty$ we have

$$H_u(\boldsymbol{x},Z) \longrightarrow \mathbf{1}\{Z(s)>0,\ s\in P\} \qquad \text{a.s.}$$

Fix a residual path satisfying the preceding bounds and having no zero passive value. If $Z(s)<0$ at some $s \in P$, evaluating at s, where $m(s)=1$, shows that $H_u(\boldsymbol{x},Z)=0$ for all large u.
Suppose instead that all passive values are positive. Near each active contact, continuity and $\boldsymbol{a}(t_i)=\boldsymbol{e}_i$ give $\boldsymbol{a}(t)^\top\boldsymbol{x} \geq x_i/2$. Since $q_i/2<\gamma_i<q_i$

$$\sup_{r\geq 0}\{K_i r^{\gamma_i}-\lambda_i u r^{q_i}\} = O\left(u^{-\gamma_i/(q_i-\gamma_i)}\right) = o(u^{-1}).$$

Thus the expression defining H_u is bounded below by $x_i/(2u)-o(u^{-1})>0$, uniformly on a sufficiently small relative neighbourhood of t_i. Near each passive contact, continuity makes Z bounded below by a positive constant; since $m \geq 1$ and $\boldsymbol{a}$ is bounded, the same expression is positive there for all large u. The complement of these relative open neighbourhoods is compact and disjoint from E. If nonempty, it has $\inf(m-1)>0$, while Z and $\boldsymbol{a}$ are bounded. Positivity follows there as well, proving the claim.
Using the conditional density f_u and its unnormalised version g_u from the standing notation, independence of Z and $\boldsymbol{X}_S$ gives

$$\mathbb{P}_u(B_u) = \frac{\displaystyle\int_{(0,\infty)^k} g_u(\boldsymbol{x})\,\mathbb{E}H_u(\boldsymbol{x},Z)\,d\boldsymbol{x}}{\displaystyle\int_{(0,\infty)^k} g_u(\boldsymbol{x})\,d\boldsymbol{x}}.$$

The pointwise convergence above and $0 \leq H_u \leq 1$ imply $\mathbb{E}H_u(\boldsymbol{x},Z) \to C$. Since

$$0 \leq g_u(\boldsymbol{x}) \leq e^{-\boldsymbol{\theta}^\top\boldsymbol{x}}, \qquad g_u(\boldsymbol{x}) \longrightarrow e^{-\boldsymbol{\theta}^\top\boldsymbol{x}}, \qquad u\to\infty$$

and every $\theta_i>0$, dominated convergence yields $\lim_{u\to\infty}\mathbb{P}_u(B_u)=C$. Since $\mathbb{P}(B_u)=\mathbb{P}_u(B_u)\mathbb{P}(A_u)$, this proves the first asymptotic relation. The same Gaussian change of variables gives

$$\mathbb{P}(A_u) = \frac{u^{-k}e^{-u^2/(2\sigma_*^2)}}{(2\pi)^{k/2}\sqrt{\det\Sigma_*}}\int_{(0,\infty)^k} g_u(\boldsymbol{x})\,d\boldsymbol{x},$$

and dominated convergence sends the integral to $\prod_{i=1}^k \theta_i^{-1}$, proving the explicit prefactor.

□

Lemma 4.3. *Fix $\beta \in (0,1]$. Let $D \subset [0,\varrho]$ be a nonempty compact set containing 0, where $\varrho \geq 0$. Let $G=\{G(x),\, x\in D\}$ be a centred GP with continuous sample paths, $G(0)=0$, and suppose that for some $0 \leq A < \infty$*

$$\mathbb{E}\{(G(x)-G(y))^2\} \leq A^2|x-y|^{2\beta}, \qquad x,y\in D.$$

Then there exists a constant $K_\beta<\infty$, depending only on β, such that uniformly in A, ϱ, and D

$$\mathbb{E}\left\{\sup_{x\in D}|G(x)|\right\} \leq K_\beta A\varrho^\beta, \qquad \sup_{x\in D}\operatorname{Var}G(x) \leq A^2\varrho^{2\beta}.$$

Proof. If $A = 0$ or $\varrho = 0$, the result is immediate from anchoring and continuity. Otherwise, the canonical metric satisfies

$$d_G(x,y) \le A|x-y|^\beta, \qquad N(D, d_G, \varepsilon) \le 1 + \varrho(A/\varepsilon)^{1/\beta}.$$

For the covering bound, a maximal $(\varepsilon/A)^{1/\beta}$-separated subset of D in the Euclidean metric is an ε-net in d_G and has at most $1+\varrho(A/\varepsilon)^{1/\beta}$ points. Since $G(0) = 0$, the variance bound follows by taking $y = 0$. Dudley's entropy bound and Gaussian symmetry give, with a universal constant K

$$\begin{aligned}\mathbb{E}\left\{\sup_{x\in D}|G(x)|\right\} &\le 2K\int_0^{A\varrho^\beta}\sqrt{\log\left(1+\varrho(A/\varepsilon)^{1/\beta}\right)}\,d\varepsilon \\ &= 2KA\varrho^\beta\int_0^1\sqrt{\log\left(1+s^{-1/\beta}\right)}\,ds \le K_\beta A\varrho^\beta.\end{aligned}$$

The integral is finite for every $\beta > 0$. □

Lemma 4.4. *Fix $\beta \in (0,1]$, $A \ge 0$, $c > 0$, and $R \in (0,\infty]$. Let G be a centred GP on $[0,R]$, interpreted as $[0,\infty)$ when $R = \infty$, with continuous sample paths, $G(0) = 0$, and*

$$\mathbb{E}\{(G(x) - G(y))^2\} \le A^2|x-y|^{2\beta}.$$

Let d be continuous on the same domain and satisfy $d(x) \ge cx^{2\beta}$. There exists a finite constant $K = K(\beta, A, c)$, independent of R, G, d, such that for $0 < L \le R$

$$\mathbb{P}\left\{\inf_{L\le x\le R}\{G(x)+d(x)\} \le 0\right\} \le KL^{-\beta}.$$

If $R = \infty$, then also $\inf_{x\ge 0}\{G(x)+d(x)\} > -\infty$ almost surely.

Proof. Put $Q_n = 2^n L$, and retain the intervals $I_n = [Q_n, 2Q_n] \cap [0,R]$ that are nonempty. Write $\varrho_n = \min\{2Q_n, R\}$. By Lemma 4.3 and Markov's inequality,

$$\begin{aligned}\mathbb{P}\left\{\sup_{0\le x\le\varrho_n}|G(x)| \ge \frac{c}{2}Q_n^{2\beta}\right\} &\le \frac{2K_\beta A\varrho_n^\beta}{cQ_n^{2\beta}} \\ &\le \frac{2^{\beta+1}K_\beta A}{c}\,Q_n^{-\beta}.\end{aligned}$$

Outside the union of these events, for every $x \in I_n$

$$G(x) + d(x) > \frac{c}{2}Q_n^{2\beta} \ge \frac{c}{2}L^{2\beta}.$$

Consequently, summing the geometric series gives the asserted estimate, with $K = 2^{\beta+1}K_\beta A/[c(1-2^{-\beta})]$. When $R = \infty$, the events $\{\inf_{x\ge L}(G(x)+d(x)) \le 0\}$ decrease as $L \to \infty$, and their probabilities tend to zero. Almost surely the process is therefore positive outside some compact interval. Continuity on that interval proves the final assertion. □

Proof of Theorem 3.5. By the standing reduction, it remains to identify the limit of $\mathbb{P}_u(B_u)$. We use the one-point overshoot convergence stated above and put

$$r_u := u^{-1/\beta}, \qquad T_u := \sup_{t\in[a,b]}\frac{-uZ(t) - u^2\{m(t)-1\}}{m(t)}.$$

Since $m \ge 1$ and $Z(b) = 0$, we have $T_u \ge 0$. Gaussian regression, compactness and continuity give the exact identity

$$B_u = \{S_u > T_u\}.$$

The rv T_u depends only on Z, so it is independent of S_u and its law is unchanged under $\mathbb{P}_u$.
Since all processes in (23) are centred Gaussian, taking $r \downarrow 0$ in the endpoint increment bound gives

$$\mathbb{E}\{(W(x) - W(y))^2\} \le C|x-y|^{2\beta}, \qquad x, y \ge 0.$$

Lemma 4.4, applied with $G = W$, $d(x) = \lambda x^{2\beta}$, and $R = \infty$, gives $M > -\infty$ almost surely. Since $W(0) = 0$, also $M \le 0$, and hence $C_M \in (0,1]$.
For every fixed $L < \infty$, the expansion (22) gives, uniformly on $0 \le x \le L$

$$u^2\{m(b - r_u x) - 1\} \longrightarrow \lambda x^{2\beta}, \qquad m(b - r_u x) \longrightarrow 1, \qquad u \to \infty.$$

Indeed, the error in the first convergence is bounded by

$$L^{2\beta} \sup_{0<h\le r_u L} \left| \frac{m(b-h)-1}{h^{2\beta}} - \lambda \right|.$$

Moreover, the finite-dimensional convergence (23) can be upgraded to weak convergence in $C[0,L]$. To see this, choose an even integer p with $p\beta > 1$. The increment bound and Gaussian moments imply, uniformly for all sufficiently large u

$$\mathbb{E}\left\{|uZ(b-r_u x) - uZ(b-r_u y)|^p\right\} \le C_p |x-y|^{p\beta}, \qquad 0 \le x, y \le L.$$

The Kolmogorov–Chentsov criterion therefore gives tightness. Setting

$$T_{u,L} := \sup_{0\le x\le L} \frac{-uZ(b-r_u x) - u^2\{m(b-r_u x)-1\}}{m(b-r_u x)},$$
$$D_L := \inf_{0\le x\le L} \{W(x) + \lambda x^{2\beta}\},$$

the continuous mapping theorem gives $T_{u,L} \Rightarrow -D_L$.
Choose $0 < r_0 < b-a$ so that the local increment bound holds and $m(b-h)-1 \ge (\lambda/2)h^{2\beta}$ for $0 \le h \le r_0$. On $[0, r_0/r_u]$, the process $G_u(x) = uZ(b-r_u x)$ satisfies

$$\mathbb{E}\{(G_u(x) - G_u(y))^2\} \le C|x-y|^{2\beta}, \qquad u^2\{m(b-r_u x)-1\} \ge \frac{\lambda}{2} x^{2\beta}.$$

Thus Lemma 4.4 bounds the probability of a nonpositive value on $Lr_u \le h \le r_0$ by $KL^{-\beta}$, uniformly in all sufficiently large u. Since $E = \{b\}$,

$$d_0 := \inf_{a\le t\le b-r_0} \{m(t)-1\} > 0.$$

As $T_{u,L} \ge 0$, an outside point can change the supremum only if its numerator is positive. Hence, for all sufficiently large u

$$\mathbb{P}\{T_u \ne T_{u,L}\} \le \mathbb{P}\left\{ \inf_{Lr_u\le h\le r_0} \left[u^2\{m(b-h)-1\} + uZ(b-h)\right] \le 0 \right\}$$
$$+ \mathbb{P}\left\{ \sup_{t\in[a,b]} |Z(t)| \ge u d_0 \right\}.$$

The last probability tends to zero by path continuity and compactness. Together with Lemma 4.4, this proves

$$\lim_{L\to\infty} \limsup_{u\to\infty} \mathbb{P}\{T_u \ne T_{u,L}\} = 0.$$

Since $D_L \downarrow M > -\infty$, we conclude that $T_u \Rightarrow -M$. Independence and the one-point overshoot limit therefore give, under $\mathbb{P}_u$

$$(S_u, T_u) \Rightarrow (\mathcal{E}, -M),$$

where $\mathcal{E}$ is independent of M and is exponential with rate $1/R(b,b)$. In particular, $\mathbb{P}\{\mathcal{E} = -M\} = 0$, and hence

$$\lim_{u\to\infty} \mathbb{P}_u(B_u) = \mathbb{P}\{\mathcal{E} > -M\} = \mathbb{E}\exp\left\{\frac{M}{R(b,b)}\right\}.$$

This proves the assertion. □

Proof of Corollary 3.6. For every fixed $L < \infty$, we have

$$\{uZ(b-x/u),\ 0 \le x \le L\} \Rightarrow \{\eta x,\ 0 \le x \le L\} \quad \text{in finite-dimensional distributions}, \qquad u \to \infty.$$

Indeed, for every fixed $x \ge 0$

$$uZ(b-x/u) = x\,\frac{Z(b-x/u)}{x/u} \longrightarrow \eta x \quad \text{in } L^2,$$

with the value at $x = 0$ understood as zero. This gives convergence of the finite-dimensional distributions. Together with the assumed local mean-square Lipschitz bound, this verifies the tangent-critical assumptions. Thus b is a tangent-critical endpoint contact with

$$\beta = 1, \qquad W(x) = \eta x, \quad x \ge 0.$$

Its variational term is

$$M_{\mathrm{sm}} := \inf_{x\ge 0} \{\eta x + \lambda x^2\} = -\frac{(\max\{-\eta, 0\})^2}{4\lambda}.$$

The conclusion of Theorem 3.5 therefore applies. Writing $s := R(b,b)$, a standard Gaussian calculation gives also when $v = 0$

$$\mathbb{E}\left\{e^{-\frac{(\max\{-\eta,0\})^2}{4\lambda s}}\right\} = \frac{1}{2} + \frac{1}{2}\mathbb{E}\exp\left\{-\frac{\eta^2}{4\lambda s}\right\} = \frac{1}{2} + \frac{1}{2}\left(1 + \frac{v^2}{2\lambda s}\right)^{-1/2}.$$

This proves the assertion. □

Appendix A. Auxiliary Results

Let $\mathcal{T}$ be a nonempty compact metric space and let $X(t)$, $t \in \mathcal{T}$ be a centred real-valued GP with almost surely continuous sample paths and continuous covariance function R. For a finite signed measure γ on $\mathcal{T}$, define the energy associated with the covariance kernel R by

$$\mathcal{E}_R(\gamma) := \iint_{\mathcal{T}^2} R(s,t)\,\gamma(ds)\gamma(dt) \geq 0.$$

For $\mu \in \mathcal{M}_1(\mathcal{T})$ define its potential and energy by

$$\phi_\mu(t) := \int_{\mathcal{T}} R(s,t)\,\mu(ds), \qquad \sigma^2(\mu) := \mathcal{E}_R(\mu).$$

We call μ an optimal measure if

$$\sigma^2(\mu) = \inf_{\nu \in \mathcal{M}_1(\mathcal{T})} \sigma^2(\nu).$$

We use the following standard optimality criterion for minimum-energy probability measures; see [4, Theorem 3.1]. Related formulations appear in [12, Theorem 4.3 and Remark 4.5], [13, Theorem 2.1], and [14, Theorem 1.1].

Lemma A.1. *An optimal measure exists. A probability measure $\mu \in \mathcal{M}_1(\mathcal{T})$ is optimal if and only if*

$$\phi_\mu(t) \geq \sigma^2(\mu), \qquad t \in \mathcal{T}.$$

For every optimal μ, equality holds on $\operatorname{supp}(\mu)$. *If further $\mathcal{E}_R(\gamma) > 0$ for every nonzero finite signed measure γ with $\gamma(\mathcal{T}) = 0$, then the optimal measure is unique.*

Corollary A.2. *Let $S = \{t_1, \ldots, t_m\} \subset \mathcal{T}$, assume that $\Sigma = (R(t_i,t_j))_{i,j=1}^m$ is invertible and set*

$$w := \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top\Sigma^{-1}\mathbf{1}}, \qquad c := \frac{1}{\mathbf{1}^\top\Sigma^{-1}\mathbf{1}}, \qquad E := \left\{t \in \mathcal{T} : \sum_{j=1}^m w_j R(t_j,t) = c\right\}.$$

If all components of w are positive and

$$\sum_{j=1}^m w_j R(t_j,t) \geq c, \qquad t \in \mathcal{T},$$

then $\mu_* := \sum_{j=1}^m w_j \delta_{t_j}$ is an optimal measure with support exactly S. If further either $\mathcal{E}_R(\gamma) > 0$ for every nonzero finite signed measure γ on $\mathcal{T}$ with $\gamma(\mathcal{T}) = 0$, or $E = S$, then μ_* is the unique optimal measure. Conversely, if an optimal measure has support exactly S, then it equals $\sum_{j=1}^m w_j \delta_{t_j}$, and w has positive components.

Proof. Because Σ is an invertible covariance matrix, it is positive definite. Hence $c > 0$, $\mathbf{1}^\top w = 1$, $\Sigma w = c\mathbf{1}$, and $\sigma^2(\mu_*) = c$. The assumed potential inequality and Lemma A.1 give optimality; the strict-energy alternative gives uniqueness.

Suppose instead that $E = S$, and let ν be another optimal measure. Since $\phi_{\mu_*} \geq c$ and both measures have energy c

$$0 \leq \mathcal{E}_R(\nu - \mu_*) = 2c - 2\int_{\mathcal{T}} \phi_{\mu_*}\,d\nu \leq 0$$

and hence $\int_{\mathcal{T}}(\phi_{\mu_*} - c)\,d\nu = 0$. Continuity and nonnegativity imply $\operatorname{supp}(\nu) \subseteq E = S$. Writing $\nu = \sum_j v_j \delta_{t_j}$, we obtain

$$0 = \mathcal{E}_R(\nu - \mu_*) = (v - w)^\top \Sigma (v - w)$$

and positive definiteness gives $v = w$.

For the converse, write the optimal measure as $\mu = \sum_j v_j \delta_{t_j}$ and put $d := \sigma^2(\mu)$. Equality of its potential and energy on its support gives $\Sigma v = d\mathbf{1}$. Together with $\mathbf{1}^\top v = 1$, this yields $d = c$ and $v = w$. Exact support gives $v_j > 0$ for every j. □

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